Danish in Emma's Path

Effort and cost analysis for a Danish/English language choice · 19 October 2026 · Based on the code at `main 5a749112` · Figures marked **estimate** are judgement, not measurement

Bottom line. The job has two halves of very different size.

- **Danish maths and app shell: medium.** The maths problems themselves don't depend on language; only what Emma says and what is shown changes. The plumbing (a language setting, Danish text and Danish audio for every line) is real work, but it is mechanical.
- **Danish reading: large, and a new course rather than a translation.** Every Word Song step is built on English phonics: 26 letters, short vowels a/o/u/i/e, sh/ch/th, and English CVC words. Danish reading needs æ, ø and å, Danish letter sounds, *lydrette ord* (words spelled as they sound) and the 120 most common words.
- **Money is small.** About **$10–30** for Danish maths, and **$40–70** in total with Danish reading (voice, AI calls, pictures). Running costs are close to $0 a month. The real cost is team time and a native Danish ear for the ear-tests.
- **Recommendation:** ship Danish maths first. In Danish mode the reading world shows as "kommer snart" (coming soon). Get the colleague's daughter onto TestFlight. Decide on Danish reading after seeing how she uses the app and after a Danish letter-sound voice test.

Danish maths + app shell (Phases 0–2)

7–11 working days

Team build time **estimate**, plus about 600 ear-test rows by a native Danish speaker. Cash: **$10–30**.

+ Danish reading world (Phase 3)

+10–20 working days

Team build time **estimate**, plus about 900 more ear-test rows. Extra cash: **$30–40**. This phase carries the highest risk: Danish letter sounds from the voice are untested.

1. What has to change: findings from the code

Area	Today (measured)	What Danish needs
Language layer	None. No i18n library, no translation calls, no language field anywhere in the app, server or audio keys.	One setting, Danish or English, which Emma, captions, menus and the map all follow.
Child's name	Emma never says "Marian" in audio (0 of about 1,500 voiced lines). But `'Marian'` is hard-coded in 9 session-request places in `mobile/src`, in about 10 parent-screen texts and in the AI planner prompt. The data model has `profile.childName`, but no screen sets it.	A name field in Parent settings, used everywhere instead of "Marian".
App text	About 1,000 lines in shared code: 548 map lines (343 not yet voiced), 226 guidance lines, 17 hub, 4 greet, 12 land names, step-name tables for 35 steps, about 45 parent texts. Screens add about 30 inline texts and 42 accessibility labels.	A Danish version of each, looked up by language. The English grammar helpers (is/are, "and", count words) need Danish counterparts.

Area	Today (measured)	What Danish needs
Emma's session lines (canon)	36 files with 2,440 lines: 1,093 distinct texts, 23,141 characters. Maths has 15 files and 404 distinct texts (10,466 chars); reading has 21 files and 709 (13,074 chars). **The app reads these English sentences with pattern-matching to build each problem** (e.g. "Nine plus two. How many?").	Swapping the text for Danish would break the problem-building. Instead, keep the English line as the structure and store Danish text and audio next to it (§2).
Bundled audio	457 MP3s in the app (10 MB), named by line id, with no language in the path.	A Danish set in its own folder (about +10 MB in the app).
Voice	ElevenLabs `eleven_v4`, voice *Lily* (British). No language code is sent. Phonics sounds are written as IPA in slashes, e.g. `/v:/`.	`eleven_v4` lists Danish as supported. Choose a Danish voice by ear-test (§4).
AI planner prompts	English-only and written for Marian ("Speaks Tagalog primarily… use only common, concrete English words"): preamble 5.6k characters, maths guide 87.6k, reading guide 69.8k.	Not needed for Danish maths if Danish sessions come only from pre-recorded sets (§2). A Danish reading world needs its own guide.
Voice-QA ear-test page	About 1,545 rows, keyed without language.	A language column, so Danish verdicts don't overwrite English ones.
Progress	Stored per device under a random device id; one child per device. Cloud backup is keyed by that id.	Nothing for a second family: a different device automatically gets separate progress.

2. Recommended design

1. **One setting.** `language: 'en' | 'da'` in Parent settings, next to the existing rows. It syncs with the cloud backup automatically, and the same screen gets the child-name field.
2. **Maths as "parallel text".** Each pre-recorded maths line keeps its English text, which the app already knows how to read, and gains a Danish text and a Danish audio clip under the same line id.
   - No maths parser needs rewriting.
   - Maths progress uses the same steps in both languages, so switching language keeps her progress.
3. **Language in every audio key.** Bundled audio, session audio and the session cache each gain a language part, e.g. `audio/da/path/…`.
4. **Danish v1 uses pre-recorded sessions only.**
   - Today the server builds a few session kinds live with AI plus voice: graduation, spaced-review and slow-fact hint sessions. In Danish v1 these fall back to the pre-recorded set, so Danish is a little less adaptive at first.
   - A live Danish path (translate, then voice at request time) can come later. It would add delay; the English live path already takes about 15 s.
5. **Danish number words.**
   - The app's number-word table is English, and Danish works differently: *enogtyve* (21) says the ones first, and *halvtreds* (50) and *firs* (80) are irregular. So a Danish table is needed.
   - Dave should also review the two-digit lines, because spoken Danish says the ones first while written numbers put the tens first.
6. **The reading world as a separate Danish set of steps**, with its own ids and progress.

- It is built from structured data, not from sentences the app has to pattern-match. That avoids the "six places must change together" trap the English tiers hit.
- Where a Danish word shares its meaning with an English one (kat/cat, sol/sun), the existing pictures can be reused.

3. Phases, effort and cost

Durations are **estimates** by comparison with this project's own record. The native port went from decision to native-only in about 6 days (8–14 Oct). Content waves 1–4 shipped in 4 days (16–19 Oct), each with a spec, 1–2 content PRs and a voice-QA baseline.

Phase	Work	Team days	Your / native-ear time
0. Decide and audition	• Questions to the colleague (§6). • Danish voice bake-off on a pick page: Lily speaking Danish against 2–3 native Danish voices from the ElevenLabs library, using real Emma lines. • An early test of about 10 Danish letter sounds.	1–2	About 1 hour of listening
1. Language plumbing	• Language setting and name field. • Text lookup by language. • Audio, canon and cache keys by language. • Language column on the voice-QA page. • Tests run in both languages. No Danish content yet.	3–5	Review only
2. Danish maths + shell	• Translate, with a fixed glossary: 404 maths lines, about 170 map and general lines, greet, hub, land names, parent texts and screen texts. • A native speaker checks the text. • Voice about 17k characters, then ear-test about 600 rows and re-take the flagged clips. • The reading world shows "kommer snart". • TestFlight build for the colleague.	3–5	About 600 ear-test rows, plus a text read-through
3. Danish reading world	• Dave researches and specs a Danish sequence: letters including æ ø å, then letter sounds, *lydrette ord*, the 120 common words, short sentences and little stories. • 6–8 content waves. • Danish letter-sound tuning, the high-risk item (§5). • 60–120 new pictures. • Ear-test about 900 rows.	10–20	About 900 ear-test rows, plus picture contact sheets

Cash costs

Item	Basis	Phases 0–2	+ Phase 3
ElevenLabs voice	• Phases 0–2: one Danish take is about 17k characters (English length; Danish may run a little longer). With re-takes and the audition, about 35–40k. • Phase 3: about 20k per take, about 50–60k with phoneme auditions. • Pricing page (read 19 Oct): Starter $6/month with 75,000 `eleven_v4` characters; Creator $22 with 275,000; overage $0.08 per 1,000.	$6–22	$22
Claude API (Haiku)	Translating about 1,100 lines and baking the Danish reading sets. Cents to a few dollars **(estimate)**.	< $5	$2–5
Pictures (OpenAI Flare)	New Danish words that can't reuse an existing picture, at $0.078 per word (current pack rate).	$0	$5–10
Apple / TestFlight	No extra cost for testers on your existing developer account.	$0	$0
EAS builds	Free plan, 15 iOS builds a month. Danish needs about 2–4 per phase.	$0	$0
Total	Running cost afterwards is about $0 a month while Danish sessions come from pre-recorded sets.	**$10–30**	**+$30–40**

Not counted: the Claude agent time that builds it, which runs on your Claude plan and wasn't measured here. Also not counted: any top-up the prepaid OpenAI and Anthropic balances need first.

Ongoing cost to keep two languages

This is the largest long-term cost. Every future maths content wave needs a Danish twin (translate, voice, ear-test), which adds roughly **30–50%** to each maths wave. If Danish reading is built, it is a second reading course to maintain alongside the English one.

4. The voice

- **Supported:** ElevenLabs' `eleven_v4` page lists Danish among its languages. The same pipeline can render Danish without changing models.
- **Which voice:** Lily is a British English voice. Speaking Danish, she may carry an English accent. A native Danish voice from the ElevenLabs voice library is the likely winner. Emma would then sound different in each language, which is fine because each child hears only one. The Phase 0 bake-off decides this, following the pattern that picked Lily (40 of 40 rows on 4 Oct).
- **Phonics sounds:** `eleven_v4` supports no SSML. The project gets clean English letter sounds by writing IPA in slashes, but that trick is proven only for English. Danish has vowel qualities and sounds such as the soft *d* and *stød* that English lacks, so expect tuning.
- **Rate limits:** the live session code assumes 3 parallel requests on Starter (measured 4 Oct). Pre-recording Danish has no time pressure, so this only matters if a live Danish path is added.

5. Risks

Risk	Why it matters	Mitigation
Danish letter sounds from the voice	English isolated sounds took 8 blend passes and 4–6 voice-QA rounds on Azure before ElevenLabs solved them. Danish has never been tried.	Test about 10 sounds in Phase 0, before committing to Phase 3.
Native ear needed	About 1,500 Danish ear-test rows in total. A non-native ear misses wrong stress and odd phrasing.	Agree up front who listens: you or the colleague.
Child-level Danish	Emma's lines are capped at very simple words. A literal translation reads like an adult.	Fixed glossary, Haiku translation with a Danish word-cap rule, and native review.
Less adaptive at first	Danish v1 skips the live AI sessions (graduation, spaced review, slow-fact hints).	Add a live Danish path later if she uses the app a lot.
Wrong target	The reading content depends on her age and whether she already reads Danish.	Answer the questions in §6 before Phase 3.

6. Getting the colleague's daughter onto the app

- **TestFlight external tester by email.** This gives him no access to your App Store Connect. The first build of each new version goes through Apple's Beta App Review, typically 24–48 hours. An internal tester would need a role on your developer team.
- **Install on his Apple ID and device**, as you do for Marian. Apple's TestFlight terms don't allow children under 13 to use TestFlight.
- **Separate progress automatically:** her device gets its own random id. Her progress, and her first name once the name field exists, will sit in your cloud-backup store. Worth telling him.
- **Same production server.** Rate limiting is per IP, so the two families don't block each other. Server changes must be released with `yarn release` before the TestFlight build that needs them.
- **Build budget:** on 18 Oct the EAS counter read 11 of 15 iOS builds for October, and that counter lags. Plan the Danish builds for November, or use Starter or a local build. TestFlight builds expire after 90 days.

> **Questions to ask the colleague before Phase 3:**
>
> - How old is she, which class is she in, and does she already read Danish?
> - What should she get from the app: maths, Danish reading, or English?
> - Which iPad and iPadOS version will she use?
>
> If the goal is *English*, keeping the existing English reading world and adding Danish help lines later is a smaller job than a Danish reading world. That route has its own voice risk, because Danish sentences would contain English words.

Sources

Code measurements (repo at `5a749112`): `public/canon/**` (line and character counts computed from `utterances[].text`); `packages/core/src/emmasPath/pathLines.ts`, `guidanceLines.ts`, `progress/lands.ts`, `progress/defaults.ts:257`; `api/_tts_elevenlabs.ts:27-32,218-222`; `api/_session.ts:161-165`; `api/_planner.ts:1599`; `packages/core/src/math/planFromServer.ts:143`; `packages/core/src/wordSong/planFromServer.ts:93`; `public/voice-qa.html:641`; `mobile/assets/audio` (457 MP3, 10 MB); `design/voice-migration-elevenlabs.md`; `design/native/content-wave-1…4.md`; `git log` since 1 Sep 2026.

Web (read 19 Oct 2026): ElevenLabs – Eleven v4 languages; ElevenLabs API pricing; Apple – TestFlight; TestFlight Terms of Service; Expo – TestFlight; Seymour, Aro & Erskine (2003), foundation literacy in European orthographies (Danish and English are slower to start reading than most); What children learn in 1st class (letter sounds and *lydrette ord*).